

Spring 2005

Problem 1 (Sp05). Let (a_n) be a sequence in $\mathbb{R}$ such that $\sum_{n=1}^{\infty} |a_{n+1} - a_n| < \infty$.

1. Prove that (a_n) is a Cauchy sequence.
2. Is the converse true?

Problem 2 (Sp05). Prove or disprove the statement: Every function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + y) = f(x) + f(y)$ for all x and y is continuous.

Problem 3 (Sp95, Sp05, Sp15). Prove that there is no one-to-one conformal map of the punctured disc $G = \{z \in \mathbb{C} \mid 0 < |z| < 1\}$ onto the annulus $A = \{z \in \mathbb{C} \mid r < |z| < R\}$.

Problem 4 (Sp05, Sp12). Suppose that V is a finite dimensional vector space over a field $\mathbb{F}$, and A, B are commuting diagonalizable linear maps from V to V . Show AB is diagonalizable.

Problem 5 (Sp05, Sp09). Let I be an open interval and let $f: I \rightarrow \mathbb{R}$ have continuous k -th derivatives everywhere on I for all $k \leq n - 1$. Let $a \in I$ be such that $f^{(k)}(a) = 0$ for $1 \leq k \leq n - 1$, and assume that $f^{(n)}(a)$ is defined and $f^{(n)}(a) > 0$. Prove that if n is even, then f has a local minimum at a , and if n is odd, then f has no local extremum at a .

Problem 6 (Sp05). For every positive integer n , define $[n]_q = q^{n-1} + q^{n-2} + \cdots + q + 1 \in \mathbb{Z}[q]$. Prove that $[1]_q [2]_q \cdots [r]_q$ divides $[k+1]_q [k+2]_q \cdots [k+r]_q$ in the polynomial ring $\mathbb{Z}[q]$, for all positive integers k and r .

Problem 7 (Sp05). Let U be a connected open subset of $\mathbb{C}$, and let f be a meromorphic function on U having at least one pole. For each $c \in U$ that is not a pole of f , let $R(c)$ equal the radius of the largest open disc centered at c on which there is some holomorphic function that agrees with f on some open neighborhood of c . Prove that $R(c)$ extends to a continuous function defined on all of U .

Problem 8 (Sp85, Sp05). Let A and B be two $n \times n$ self-adjoint, i.e., Hermitian matrices over $\mathbb{C}$ and assume A is positive definite. Prove that all eigenvalues of AB are real.

Problem 9 (Sp05). Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be an infinitely differentiable function that is zero outside some bounded subset of $\mathbb{R}^2$. Prove that

$$\lim_{\varepsilon \rightarrow 0} \iint_{x^2 + y^2 \geq \varepsilon^2} \frac{f(x, y)}{(x + iy)^3} dx dy$$

exists.

Problem 10 (Sp05). Let G be a finite group. Suppose $ab = ba$ holds whenever $a, b \in G$ have prime power order. Prove that G is abelian.

Problem 11 (Sp93, Sp05). Let a be a complex number and ε a positive number. Prove that the function $f(z) = \sin z + 1/z - a$ has infinitely many zeros in the strip $|\Im z| < \varepsilon$.

Problem 12 (Sp05). Let $M_{n \times n}(\mathbb{F})$ be the ring of $n \times n$ matrices over a field $\mathbb{F}$. Prove that for every $A \in M_{n \times n}(\mathbb{F})$ there exists $X \in M_{n \times n}(\mathbb{F})$ such that $AXA = A$.

Problem 13 (Sp05). Let D be a subset of $\mathbb{R}$, and let $f: D \rightarrow \mathbb{R}$ be a function with graph G_f . Prove that if G_f is compact, then f is continuous. Note: See also ????

Problem 14 (Sp05, Sp09). Let $\mathbb{Q}(x)$ be the field of rational functions of one variable over $\mathbb{Q}$. Let $i: \mathbb{Q}(x) \rightarrow \mathbb{Q}(x)$ be the unique field automorphism such that $i(x) = x^{-1}$. Prove that the subfield of elements fixed by i is equal to $\mathbb{Q}(x + x^{-1})$.

Problem 15 (Sp05). Evaluate the integral $\int_0^\infty \frac{x \sin x}{x^2 + a^2} dx$, where $a > 0$.

Problem 16 (Sp05). Let $\mathbb{F}_p$ be the field of p elements. Let $\text{SL}_2(\mathbb{F}_p)$ be the group of 2×2 matrices over $\mathbb{F}_p$ of determinant 1. Let G be a normal subgroup of $\text{SL}_2(\mathbb{F}_p)$. Suppose G contains a nonidentity element γ that fixes a nonzero vector v . Show that any $\gamma' \in \text{SL}_2(\mathbb{F}_p)$ that fixes a nonzero vector v' belongs to G .

Problem 17 (Sp05). Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable on $\mathbb{R}$. Suppose that $f(0) = 0$, and that $|f'(x)| \leq |f(x)|$ for all $x \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \in \mathbb{R}$.

Problem 18 (Sp05). 1. Prove that if $n > 0$ is even, there does not exist $f(x) \in \mathbb{R}[x]$ such that $f(x)^2 - x$ is divisible by $x^n - 1$.

2. For odd $n > 0$, find the number of $f(x) \in \mathbb{R}[x]$ of degree less than n such that $f(x)^2 - x$ is divisible by $x^n - 1$.